Political Parties and Personalities

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POLITICAL PARTIES AND PERSONALITIES

In this study we want to find out if there are correlations and significant variables between traits of personalities and the Political Parties chosen by people answering the question “Cuál de los siguientes partidos políticos cree que apoya la lucha contra el cambio climático?

1) Correlation between political parties and personality

To start I run a Correlation Matrix to see if there are significant correlations between personalities traits and the political parties. This could be useful to have a general idea on how these data are interacting in order to move on with further investigations.

```
#to start we see a general correlation matrix to see wich are the parties and the traits that are more correlated

y1<-y[c(83:90, 94,95,99, 219:242)]
y1<-na.omit(y1)

mat0<-cor(y1)
mat1<-mat0[-c(1:11), -c(12:35)]
mat1<-as.data.frame(mat1)
knitr::kable(mat1)
```

	P43R43.1OptionVOX	P43R43.2OptionPODEM	P43R43.3OptionIZQUI	P43R43.4OptionPNV	P43R43.5OptionPP	P43R43.6OptionMÃ.SPA	P43R43.7OptionBILDU	P4
Sincerity.1	-0.0322765	0.0158126	-0.0040611	0.0330114	0.0292075	0.0684516	0.0539593	
Fairness.2	-0.0223100	0.0152008	0.0389240	-0.0008508	0.0316903	0.0423010	0.0533566	
GreedAvoidance.3	-0.0167129	0.0167299	0.0669393	0.0325603	-0.0067452	0.0461706	-0.0023203	
Modesty.4	-0.0972772	0.0496723	0.0339417	-0.0254714	-0.0389195	0.0892908	0.0116152	
Fearfulness.5	0.0094571	-0.0008058	-0.0041185	-0.0526731	0.0004078	-0.0605601	-0.1434285	
Anxiety.6	0.0063919	-0.0344252	0.0755508	-0.0116003	0.0172183	-0.0504766	-0.0249657	
Dependence.7	-0.0733248	-0.0665486	0.0082359	-0.0225731	-0.0396128	-0.0831894	-0.1144354	
Sentimentality.8	-0.0478848	-0.0138289	0.0354209	-0.0114803	-0.0170719	-0.0499538	-0.0727932	
SocialSelf.Esteem.9	-0.0288774	-0.0483040	-0.0302903	0.0311253	-0.0069005	0.0086843	0.0444781	
SocialBoldness.10	0.0793003	0.0266957	0.0560271	0.0192707	-0.0004668	0.0340606	0.0875223	
Sociability.11	0.0118816	0.0646304	0.0530224	0.0008984	-0.0074902	0.0472348	-0.0280525	
Liveliness.12	0.0179925	0.0004656	-0.0377854	0.0535187	0.0191116	-0.0131023	0.0042287	
Forgiveness.13	-0.0127056	0.0492780	0.0403625	0.0559069	0.0459225	0.0050443	0.0278220	
Gentleness.14	-0.0876064	0.0044519	-0.0096657	0.0354805	-0.0571269	0.0227921	-0.0311274	
Flexibility.15	-0.0251095	-0.1038870	-0.0969468	-0.0155710	0.0184304	-0.1499290	-0.0634935	
Patience.16	-0.0807188	0.1046738	0.0331362	0.0645202	-0.0523702	0.1138305	0.0524292	
Organization.17	0.0082563	-0.0052218	-0.0421112	0.0119396	-0.0292037	0.0096133	0.0483528	
Diligence.18	0.0140910	-0.0857656	-0.0282279	0.0272854	0.0328394	-0.0773415	0.0323801	
Perfectionism.19	-0.0060693	-0.0800614	-0.0473960	0.0530908	0.0138034	-0.0574290	-0.0484843	
Prudence.20	-0.0977124	0.0833199	0.1172584	-0.0088973	-0.0817171	0.1026784	0.0848727	
AestheticApretiation.21	-0.0987675	0.0530385	0.0240816	-0.0065661	-0.0699098	0.0539796	0.0667784	
Inquisitiveness22	-0.0234575	0.0720764	0.1424406	0.0431835	0.0122562	0.0470125	0.0525441	
Creativity.23	0.0337333	-0.0465071	0.0119918	0.0120448	0.0168094	-0.0497798	-0.0828300	
Unconventionality.24	-0.0741652	0.1048042	0.1413293	0.0084869	-0.0894294	0.0764502	0.0848329	

Conclusion 1

we can recognize that there are traits of personality more connected to the choose of political party that people believe is supporting the fight against climate change. However this correlation most of the time it’s not strong

2) Probability of choosing a party fighting against climate change due to personality traits

Now i want to see if these correlations in traits of personality are significant or Not in voting a specific political parties. In particular i want to see which traits of personality give more chances to be chosen as a party that is fighting against climate change.

To pursue this goal i will run a logistical regression for each political parties, doing this i will have a statistical model that will show the traits most significant among the parties and it will give us a probability of choosing that party. These type of regression it’s also usually used when the dependent variable is Dicotomic and i’m in this case.

- The dependent variable is the Political Party, for Y=0 we have that the individual didn’t choose the party as a party that is fighting against climate change, for Y=1 we have that the individual chose the party as a party that is fighting against climate change.
- The independent variables are the traits of personality.

1. At first i will run a logistic regression among all the variables to find a general logistic model, then i will use the backward regression on that model to select just the significant variables. The Backward selection algorithm starts from model that includes all possible regressors available and proceeds by eliminating one variable at a time taking into account the increase in residual deviance.
2. I check if the logistic model is significant calculating the PSEUDO-R2, if the p_value is low our model is significant if high we can’t consider our model a good model to continue the investigation
3. I Test the quality of my model creating a new data frame of predicted value, then I create a matrix with the positive true values and the false positive values to calculate a percentage of accuracy of my model. In this phase i will use the value of 0.5 for the predicted values (standard value).
4. I improve the model choosing a more accurate value for the predicted values. I do this creating a graph in which on the y i have the true positive rate and on the x the false positive rate. To create the new matrix i will use as a value the one that gives me the opportunity to increase the true positive rate avoiding to increase more the false positive rate.
5. I create the graph of the predicted probability to graphically visualize my model. The observation will have two colors, green for “yes” (the person chose the party) and red for “no” (the person didn’t choose the party), these observation are distributed second the probability that our model gave to them. For low probability we expect more “red” observation, for high probability we expect more “green” observation. The more our model will be accurate the more our graph will show a clear separation between green and red observation.
6. i repeat that process for all the political parties

VOX

```
#summary Backward regression

summary(model2VOX) #The model considers significant 21 variables
```

```
##
## Call:
## glm(formula = training$P43R43.1OptionVOX ~ Sincerity.1 + Fairness.2 +
##     GreedAvoidance.3 + Modesty.4 + Fearfulness.5 + Anxiety.6 +
##     Dependence.7 + SocialSelf.Esteem.9 + SocialBoldness.10 +
##     Sociability.11 + Liveliness.12 + Gentleness.14 + Flexibility.15 +
##     Organization.17 + Perfectionism.19 + Prudence.20 + AestheticApretiation.21 +
##     Inquisitiveness22 + Creativity.23 + Unconventionality.24,
##     family = "binomial", data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.2151  -0.4572  -0.3212  -0.1950   2.6782
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)             -2.74345    0.06154 -44.582  < 2e-16 ***
## Sincerity.1              0.20507    0.09962   2.058 0.039544 *
## Fairness.2              -0.31636    0.08173  -3.871 0.000109 ***
## GreedAvoidance.3         0.19386    0.08192   2.367 0.017956 *
## Modesty.4               -0.78453    0.10729  -7.312 2.63e-13 ***
## Fearfulness.5            0.22320    0.08622   2.589 0.009633 **
## Anxiety.6                1.16265    0.10859  10.707  < 2e-16 ***
## Dependence.7            -1.04270    0.09974 -10.454  < 2e-16 ***
## SocialSelf.Esteem.9     -0.24269    0.09216  -2.633 0.008456 **
## SocialBoldness.10        0.18264    0.07287   2.506 0.012202 *
## Sociability.11          -0.13255    0.08365  -1.585 0.113065
## Liveliness.12            0.40931    0.08561   4.781 1.74e-06 ***
## Gentleness.14           -0.72359    0.07547  -9.588  < 2e-16 ***
## Flexibility.15           0.28715    0.08531   3.366 0.000763 ***
## Organization.17          0.47210    0.09671   4.882 1.05e-06 ***
## Perfectionism.19        -0.88043    0.10848  -8.116 4.82e-16 ***
## Prudence.20             -0.48508    0.09835  -4.932 8.13e-07 ***
## AestheticApretiation.21 -0.49980    0.08280  -6.037 1.57e-09 ***
## Inquisitiveness22       -0.82112    0.09632  -8.525  < 2e-16 ***
## Creativity.23            0.48505    0.07197   6.740 1.58e-11 ***
## Unconventionality.24    -0.65408    0.09974  -6.558 5.47e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3819.7  on 6219  degrees of freedom
## Residual deviance: 3233.1  on 6199  degrees of freedom
## AIC: 3275.1
##
## Number of Fisher Scoring iterations: 6
```

- The model considers significant 21 variables

```
#### PSEUDO R2 ####

ll.dev<-model2VOX$null.deviance/(-2)
ll.likehood<-model2VOX$deviance/(-2)

R<-ll.dev/ll.likehood

pvalue<- 1-pchisq(2*(ll.likehood-ll.dev), df=(length(model2VOX$coefficients)-1)) #significativo
pvalue
```

```
## [1] 0
```

- Our model is significant

```
####previsioni ####

resVOX<-predict(model2VOX, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano and
ati in escursione
                                                       #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.1OptionVOX, Predictedvalue= resVOX>0.5))
```

	FALSE	TRUE
no	1404	3
si	126	22

```
(1402+21)/(1402+21+127+5) #modello ha un'accuratezza del 91,51%%
```

```
## [1] 0.9151125
```

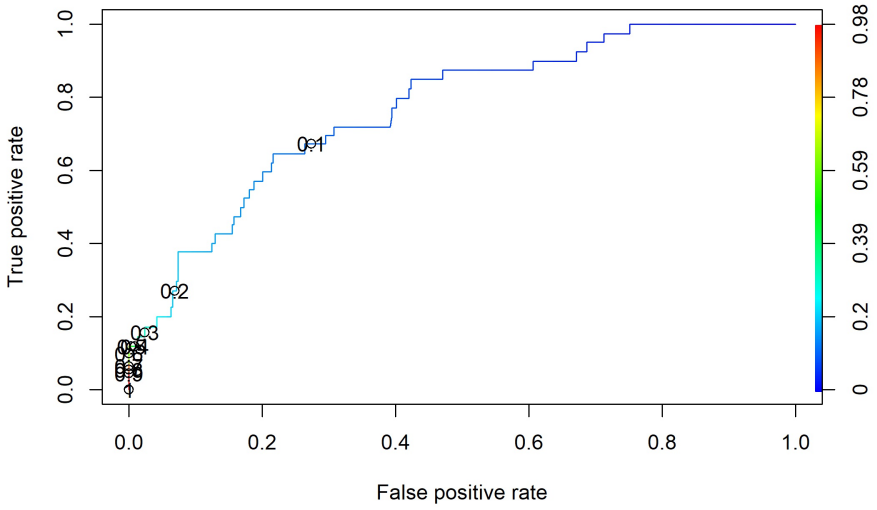
- The matrix show that model has an accuracy of 91.51%, more in details the matrix shows that the model is very good in predicting when a person not choose VOX as a political party that is fighting against climate change, but it fails in predicting when a person choose VOX. Now i try to improve the prevision for the TRUE values

```
####migliorare la previsione####

res2VOX<-predict(model2VOX, training, type = "response") #predetto su dati training

ROCRPred.VOX=prediction(res2VOX, training$P43R43.1OptionVOX)
ROCRPref.VOX<-performance(ROCRPred.VOX, "tpr", "fpr")

plot(ROCRPref.VOX, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.2 come valore
```



- This graph suggest that i should use the values of 0.2 instead of 0.5, this means that our model will consider the values above 0.2 as True and all the values under 0.2 as FALSE.

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.1OptionVOX, Predictedvalue= resVOX>0.2))
```

	FALSE	TRUE
no	1323	84
si	105	43

```
(1319+41)/(1319+41+107+88) #accuratezza del 87,46%
```

```
## [1] 0.8745981
```

- Using the new value we can see that our model lose a little percentage of accuracy, however now the model can predict a little bit better when a person chose VOX as a party that is fighting against climate change. Now i want to graphically visualize my model.

```
####grafico delle probabilità predette ####

#create new table with data i need

dataVOX<-data.frame( prob.of.vote= model2VOX$fitted.values,
                     vote = training$P43R43.1OptionVOX) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

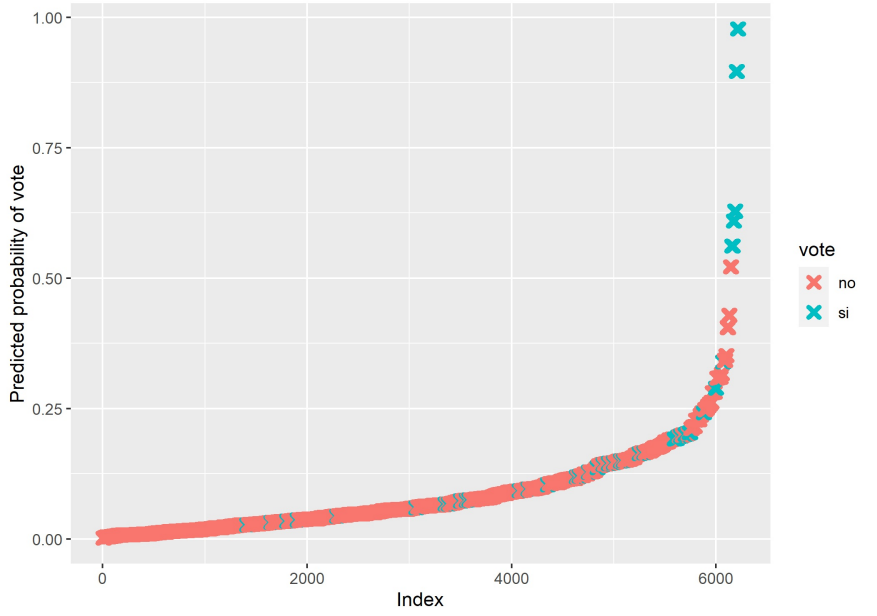
dataVOX<-dataVOX[order(dataVOX$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataVOX$rank<-1:nrow(dataVOX)

#grafico

ggplot( data=dataVOX ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.2, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability.

PODEMOS

```
#summary backward

summary(model2PODEMOS)
```

```
##
## Call:
## glm(formula = training$P43R43.2OptionPODEM ~ Fairness.2 + GreedAvoidance.3 +
##     Modesty.4 + Fearfulness.5 + Dependence.7 + Sentimentality.8 +
##     Sociability.11 + Forgiveness.13 + Flexibility.15 + Patience.16 +
##     Organization.17 + Diligence.18 + Perfectionism.19 + Prudence.20 +
##     Inquisitiveness22 + Creativity.23 + Unconventionality.24,
##     family = "binomial", data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.1058  -0.9727  -0.6685   1.0966   2.5956
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)          -0.36801    0.02830 -13.005  < 2e-16 ***
## Fairness.2           -0.18672    0.04726  -3.951 7.77e-05 ***
## GreedAvoidance.3      0.12373    0.04517   2.740  0.00615 **
## Modesty.4             0.15594    0.04928   3.164  0.00155 **
## Fearfulness.5         0.45217    0.04988   9.064  < 2e-16 ***
## Dependence.7         -0.44901    0.04989  -9.001  < 2e-16 ***
## Sentimentality.8      0.10827    0.04727   2.290  0.02199 *
## Sociability.11        0.51181    0.05024  10.187  < 2e-16 ***
## Forgiveness.13        0.13599    0.04308   3.157  0.00159 **
## Flexibility.15       -0.50145    0.04728 -10.605  < 2e-16 ***
## Patience.16           0.42048    0.04416   9.521  < 2e-16 ***
## Organization.17      -0.12109    0.05572  -2.173  0.02977 *
## Diligence.18         -0.23715    0.04978  -4.764 1.90e-06 ***
## Perfectionism.19     -0.44206    0.05304  -8.335  < 2e-16 ***
## Prudence.20           0.75130    0.06020  12.480  < 2e-16 ***
## Inquisitiveness22     0.42082    0.05130   8.204 2.33e-16 ***
## Creativity.23        -0.19899    0.03996  -4.979 6.39e-07 ***
## Unconventionality.24  0.45031    0.05199   8.661  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 8480.8  on 6219  degrees of freedom
## Residual deviance: 7591.6  on 6202  degrees of freedom
## AIC: 7627.6
##
## Number of Fisher Scoring iterations: 4
```

- the significant variables are 20

```
#### PSEUDO R2 ####

ll.dev<-model2PODEMOS$null.deviance/(-2)
ll.likehood<-model2PODEMOS$deviance/(-2)

R<-ll.dev/ll.likehood

pvalue<- 1-pchisq(2*(ll.likehood-ll.dev), df=(length(model2PODEMOS$coefficients)-1)) #significativo
pvalue
```

```
## [1] 0
```

- the model is significant

```
####previsioni ####

resPODEMOS<-predict(model2PODEMOS, testing, type = "response") #abbiamo le prob per le osservazioni testing che s
iano andati in escursione
                                                #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.2OptionPODEM, Predictedvalue= resPODEMOS>0.5))
```

	FALSE	TRUE
no	699	186
si	344	326

```
(718+309)/(718+309+339+189) #modello ha un'accuratezza del 66,05%
```

```
## [1] 0.6604502
```

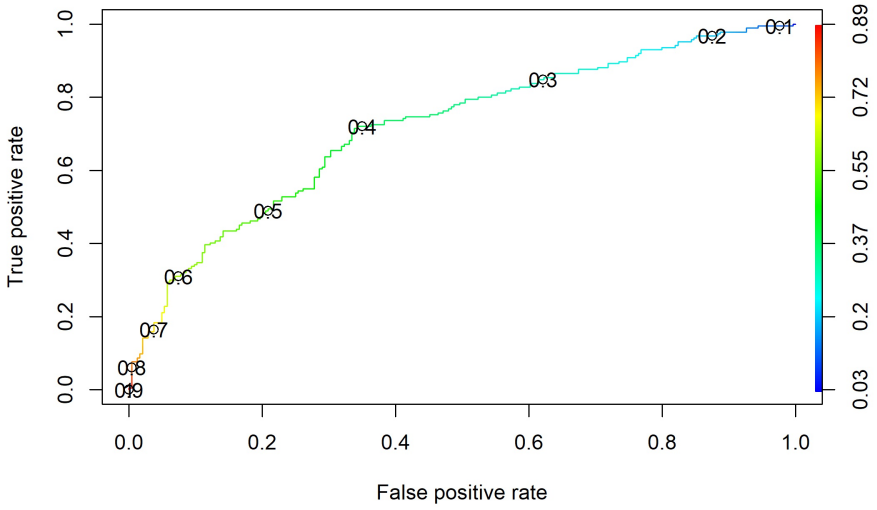
- The matrix show that model has an accuracy of 66.05%, more in details the matrix shows that the model is good in predicting when a person not choose PODEMOS as a political party that is fighting against climate change, but it fails in predicting when a person choose PODEOS (it predicts well just the 50% of times). Now i try to improve the prevision for the TRUE values.

```
####migliorare la previsione####

res2PODEMOS<-predict(model2PODEMOS, training, type = "response") #predetto su dati training

ROCRPred.PODEMOS=prediction(res2PODEMOS, training$P43R43.2OptionPODEM)
ROCRPref.PODEMOS<-performance(ROCRPred.PODEMOS, "tpr", "fpr")

plot(ROCRPref.PODEMOS, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.6 come valore
```



- This graph suggest that i should use the values of 0.6 instead of 0.5, this means that our model will consider the values above 0.6 as True and all the values under 0.6 as FALSE

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.2OptionPODEM, Predictedvalue= resPODEMOS>0.6))
```

	FALSE	TRUE
no	822	63
si	464	206

```
(836+199)/(836+199+449+71) #accuratezza del 66,56%
```

```
## [1] 0.6655949
```

- Using the new value we can see that our model gain a little percentage of accuracy, however now the model can predict a little bit better when a person didn't choose PODEMOS as a party that is fighting against climate change and predict worse when people choose PODEMOS. Now i want to graphically visualize my model. we can conclude that the values 0.5 was better.

```
####grafico delle probabilità predette ####

#create new table with data i need

dataPODEMOS<-data.frame( prob.of.vote= model2PODEMOS$fitted.values,
                 vote = training$P43R43.2OptionPODEM) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

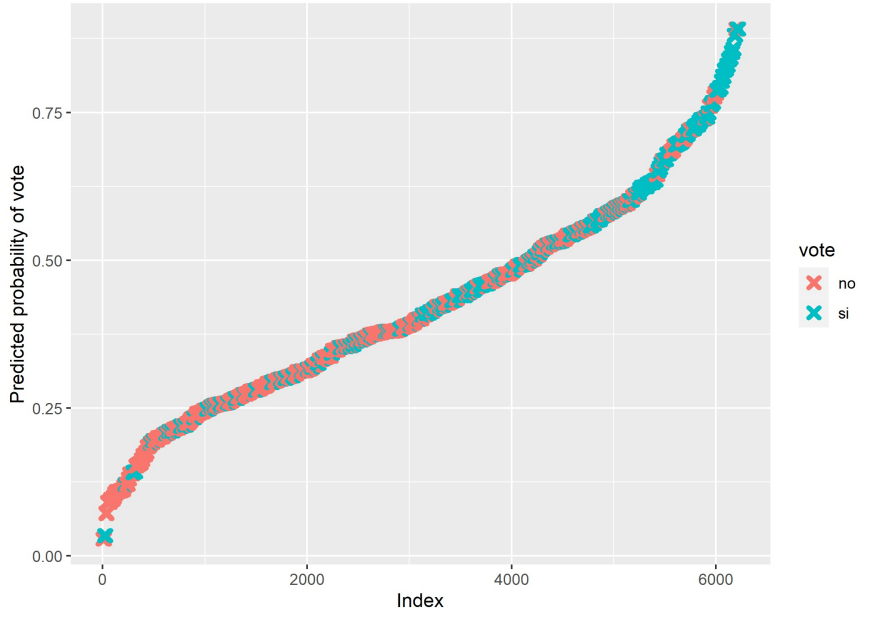
dataPODEMOS<-dataPODEMOS[order(dataPODEMOS$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataPODEMOS$rank<-1:nrow(dataPODEMOS)

#grafico

ggplot( data=dataPODEMOS ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.6, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability. In general this model it's not very good because it has an accuracy of 66.56%.

IZQUIERDA UNITA

```
#summary backward

summary(model2IZQUI)
```

```
##
## Call:
## glm(formula = training$P43R43.3OptionIZQUI ~ Sincerity.1 + GreedAvoidance.3 +
##     Modesty.4 + Fearfulness.5 + Anxiety.6 + Dependence.7 + SocialSelf.Esteem.9 +
##     SocialBoldness.10 + Sociability.11 + Liveliness.12 + Forgiveness.13 +
##     Gentleness.14 + Flexibility.15 + Patience.16 + Organization.17 +
##     Diligence.18 + Perfectionism.19 + Prudence.20 + AestheticApretiation.21 +
##     Inquisitiveness22 + Creativity.23 + Unconventionality.24,
##     family = "binomial", data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.1263  -0.9237  -0.6251   1.0542   2.1100
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)             -0.66931    0.03015 -22.201  < 2e-16 ***
## Sincerity.1             -0.19756    0.05779  -3.419 0.000629 ***
## GreedAvoidance.3         0.28638    0.04912   5.830 5.53e-09 ***
## Modesty.4                0.18959    0.06428   2.949 0.003185 **
## Fearfulness.5            0.19734    0.05312   3.715 0.000203 ***
## Anxiety.6                0.43443    0.07156   6.071 1.27e-09 ***
## Dependence.7            -0.19341    0.05534  -3.495 0.000475 ***
## SocialSelf.Esteem.9      0.19477    0.05821   3.346 0.000820 ***
## SocialBoldness.10        0.17318    0.04695   3.689 0.000225 ***
## Sociability.11           0.33868    0.05641   6.003 1.93e-09 ***
## Liveliness.12           -0.19454    0.04985  -3.903 9.51e-05 ***
## Forgiveness.13           0.09013    0.04754   1.896 0.057976 .
## Gentleness.14            0.11434    0.04842   2.361 0.018202 *
## Flexibility.15          -0.56064    0.05529 -10.139  < 2e-16 ***
## Patience.16              0.17958    0.05196   3.457 0.000547 ***
## Organization.17         -0.32598    0.05831  -5.590 2.27e-08 ***
## Diligence.18            -0.43023    0.05792  -7.428 1.10e-13 ***
## Perfectionism.19        -0.32484    0.06215  -5.227 1.72e-07 ***
## Prudence.20              0.80606    0.06388  12.617  < 2e-16 ***
## AestheticApretiation.21  0.12474    0.05798   2.151 0.031440 *
## Inquisitiveness22        0.47312    0.06282   7.531 5.02e-14 ***
## Creativity.23           -0.10899    0.04427  -2.462 0.013824 *
## Unconventionality.24     0.54997    0.05994   9.176  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 8181.9  on 6219  degrees of freedom
## Residual deviance: 7202.0  on 6197  degrees of freedom
## AIC: 7248
##
## Number of Fisher Scoring iterations: 4
```

- the model give us 22 significant variables

```
#### PSEUDO R2 ####

ll.dev<-model2IZQUI$null.deviance/(-2)
ll.likehood<-model2IZQUI$deviance/(-2)

R<-ll.dev/ll.likehood

pvalue<- 1-pchisq(2*(ll.likehood-ll.dev), df=(length(model2IZQUI$coefficients)-1)) #significativo
pvalue
```

```
## [1] 0
```

- the model is significant

```
####previsioni ####

resIZQUI<-predict(model2IZQUI, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano
andati in escursione
                                                    #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.3OptionIZQUI, Predictedvalue= resIZQUI>0.5))
```

	FALSE	TRUE
no	862	136
si	316	241

```
(830+253)/(830+253+329+143) #modello ha un'accuratezza del 69,65%
```

```
## [1] 0.696463
```

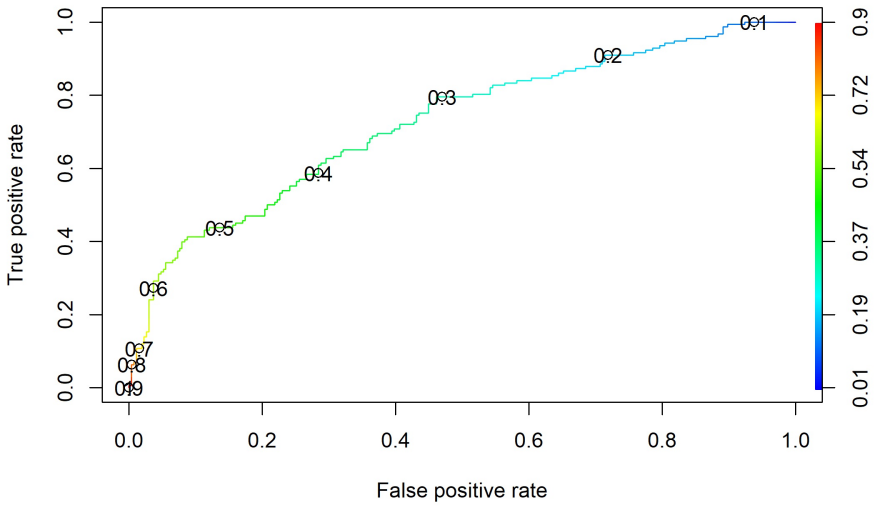
- The matrix show that model has an accuracy of 69.64%, more in details the matrix shows that the model is good in predicting when a person not choose IZQUI as a political party that is fighting against climate change, but it fails in predicting when a person choose IZQUI. Now i try to improve the prevision for the TRUE values.

```
####migliorare la previsione####

res2IZQUI<-predict(model2IZQUI, training, type = "response") #predetto su dati training

ROCRPred.IZQUI=prediction(res2IZQUI, training$P43R43.3OptionIZQUI)
ROCRPref.IZQUI<-performance(ROCRPred.IZQUI, "tpr", "fpr")

plot(ROCRPref.IZQUI, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.6 come valore
```



- This graph suggest that i should use the values of 0.6 instead of 0.5, this means that our model will consider the values above 0.6 as True and all the values under 0.6 as FALSE

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.3OptionIZQUI, Predictedvalue= resIZQUI>0.6))
```

	FALSE	TRUE
no	964	34
si	406	151

```
(936+163)/(936+163+419+37) #accuratezza del 70,67%
```

```
## [1] 0.7067524
```

- Using the new value we can see that our model gain a little percentage of accuracy, however now the model can predict a little bit better when a person didn't choose IZQUI as a party that is fighting against climate change and predict worse when people choose IZQUI. Now i want to graphically visualize my model. we can conclude that the values 0.5 was better.

```
####grafico delle probabilità predette ####

#create new table with data i need

dataIZQUI<-data.frame( prob.of.vote= model2IZQUI$fitted.values,
                  vote = training$P43R43.3OptionIZQUI) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

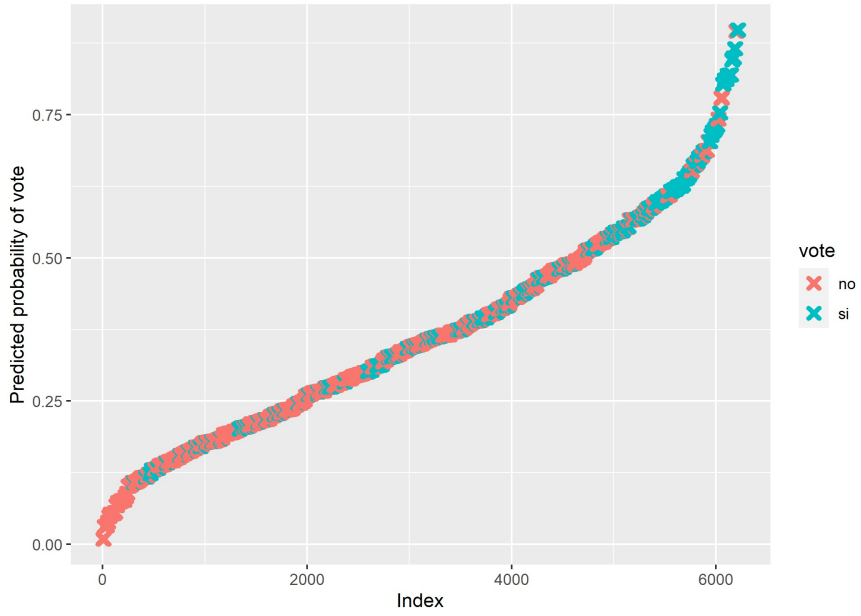
dataIZQUI<-dataIZQUI[order(dataIZQUI$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataIZQUI$rank<-1:nrow(dataIZQUI)

#grafico

ggplot( data=dataIZQUI ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.6, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability.

PNV

```
#summary backward
summary(model2PNV)
```

```
##
## Call:
## glm(formula = training$P43R43.4OptionPNV ~ Sincerity.1 + Fairness.2 +
##     Modesty.4 + Fearfulness.5 + Anxiety.6 + Dependence.7 + Sentimentality.8 +
##     SocialSelf.Esteem.9 + Sociability.11 + Liveliness.12 + Patience.16 +
##     Diligence.18 + Perfectionism.19 + Prudence.20 + AestheticApretiation.21 +
##     Inquisitiveness22 + Creativity.23 + Unconventionality.24,
##     family = "binomial", data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -0.9133  -0.3910  -0.3091  -0.2206   2.8548
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)             -3.01703    0.06724 -44.872  < 2e-16 ***
## Sincerity.1              0.58524    0.11395   5.136 2.81e-07 ***
## Fairness.2              -0.29571    0.10284  -2.876 0.004033 **
## Modesty.4               -0.53448    0.11880  -4.499 6.83e-06 ***
## Fearfulness.5           -0.25874    0.09997  -2.588 0.009653 **
## Anxiety.6               -0.25763    0.12955  -1.989 0.046733 *
## Dependence.7             0.40090    0.10312   3.888 0.000101 ***
## Sentimentality.8        -0.39391    0.10513  -3.747 0.000179 ***
## SocialSelf.Esteem.9      0.38334    0.10377   3.694 0.000221 ***
## Sociability.11           0.24380    0.10069   2.421 0.015468 *
## Liveliness.12            0.13763    0.08407   1.637 0.101625
## Patience.16              0.66832    0.08842   7.558 4.08e-14 ***
## Diligence.18             0.41862    0.10655   3.929 8.53e-05 ***
## Perfectionism.19         0.51407    0.12266   4.191 2.78e-05 ***
## Prudence.20             -0.62711    0.11993  -5.229 1.70e-07 ***
## AestheticApretiation.21  0.21956    0.11888   1.847 0.064757 .
## Inquisitiveness22        0.41546    0.10570   3.931 8.47e-05 ***
## Creativity.23           -0.21105    0.08114  -2.601 0.009295 **
## Unconventionality.24     0.69503    0.11320   6.140 8.25e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2860.8  on 6219  degrees of freedom
## Residual deviance: 2649.6  on 6201  degrees of freedom
## AIC: 2687.6
##
## Number of Fisher Scoring iterations: 6
```

- there are 19 significant variables

```
#### PSEUDO R2 ####

ll.dev<-model2PNV$null.deviance/(-2)
ll.likehood<-model2PNV$deviance/(-2)

R<-ll.dev/ll.likehood

pvalue<- 1-pchisq(2*(ll.likehood-ll.dev), df=(length(model2PNV$coefficients)-1)) #significativo
pvalue
```

```
## [1] 0
```

- the model is significant

```
####previsioni ####

resPNV<-predict(model2PNV, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano and
ati in escursione
                                                       #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.4OptionPNV, Predictedvalue= resPNV>0.5)) #a 0,5 non predice valori
TRUE
```

	FALSE
no	1467
si	88

```
(1459)/(1459+96) #93.82%
```

```
## [1] 0.9382637
```

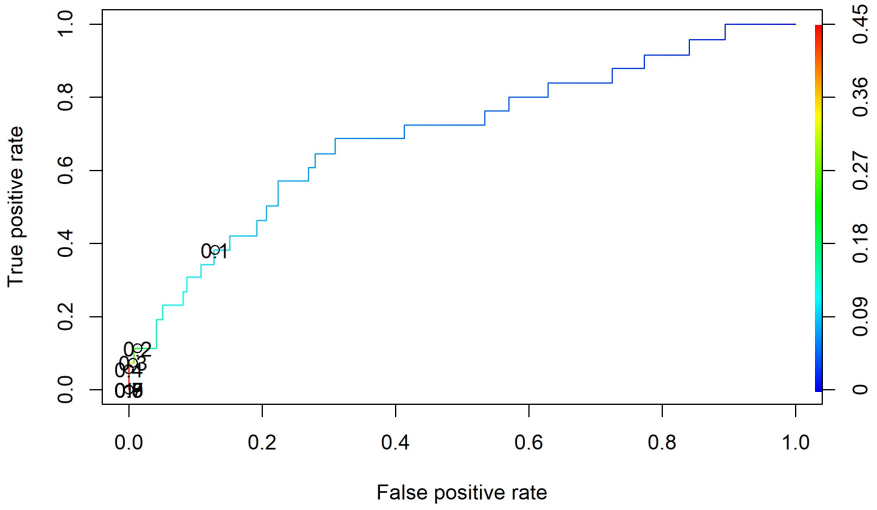
- The matrix show that model has an accuracy of 93.83%, more in details the matrix shows that the model perfectly predict when a person not choose PNV as a political party that is fighting against climate change, but it fails completly in predicting when a person choose PNV. Now i try to improve the prevision for the TRUE values.

```
####migliorare la previsione####

res2PNV<-predict(model2PNV, training, type = "response") #predetto su dati training

ROCRPred.PNV=prediction(res2PNV, training$P43R43.4OptionPNV)
ROCRPref.PNV<-performance(ROCRPred.PNV, "tpr", "fpr")

plot(ROCRPref.PNV, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.2 come valore
```



- This graph suggest that i should use the values of 0.2 instead of 0.5, this means that our model will consider the values above 0.2 as True and all the values under 0.2 as FALSE

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.4OptionPNV, Predictedvalue= resPNV>0.2))
```

	FALSE	TRUE
no	1448	19
si	77	11

```
(1446+11)/(1446+11+18+80) #accuratezza del 93,70%
```

```
## [1] 0.9369775
```

- Using the new value we can see that our model lose a little percentage of accuracy, however now the model can predict a little bit better when a person chose PNV as a party that is fighting against climate change. The improvment it's very low. Now i want to graphically visualize my model.

```
####grafico delle probabilità predette ####

#create new table with data i need

dataPNV<-data.frame( prob.of.vote= modelPNV$fitted.values,
                     vote = training$P43R43.4OptionPNV) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

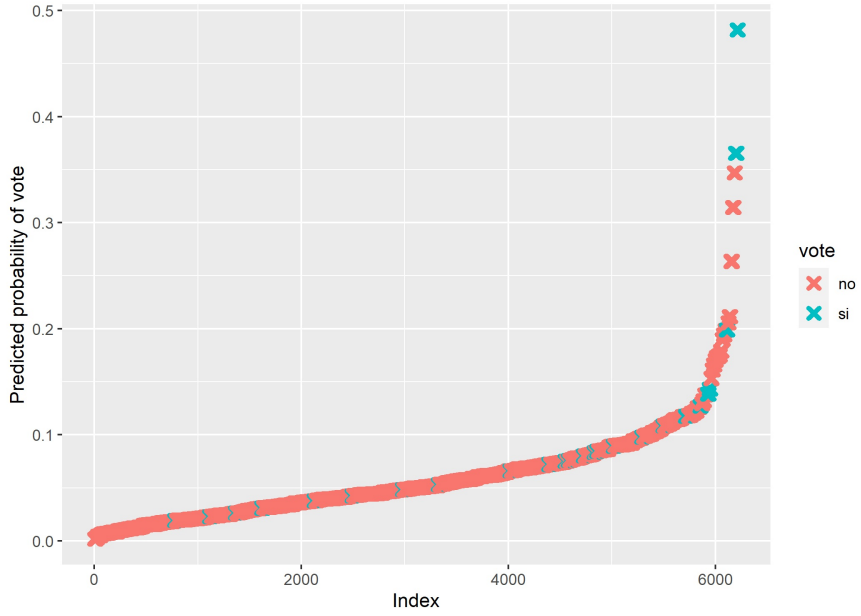
dataPNV<-dataPNV[order(dataPNV$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataPNV$rank<-1:nrow(dataPNV)

#grafico

ggplot( data=dataPNV ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.2, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability. These model can't be used to predict "green observation" but it's extremely precise in predicting when a person don't choose PNV as a party that is fighting against climate change.

PP

```
#summary backward
summary(modelPP)
```

```
##
## Call:
## glm(formula = training$P43R43.5OptionPP ~ ., family = "binomial",
##     data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.2702  -0.5348  -0.4242  -0.3048   2.5301
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)             -2.17959    0.04602 -47.363  < 2e-16 ***
## Sincerity.1              0.35565    0.08668   4.103 4.07e-05 ***
## Fairness.2               0.15059    0.07590   1.984  0.04727 *
## GreedAvoidance.3        -0.34033    0.07395  -4.602 4.18e-06 ***
## Modesty.4               -0.31270    0.09632  -3.246  0.00117 **
## Fearfulness.5           -0.07122    0.07574  -0.940  0.34707
## Anxiety.6                0.11375    0.10873   1.046  0.29547
## Dependence.7            -0.26427    0.08266  -3.197  0.00139 **
## Sentimentality.8         0.02108    0.08057   0.262  0.79358
## SocialSelf.Esteem.9     -0.21965    0.08609  -2.551  0.01073 *
## SocialBoldness.10       -0.31056    0.06926  -4.484 7.34e-06 ***
## Sociability.11          -0.04927    0.07900  -0.624  0.53279
## Liveliness.12            0.23474    0.07163   3.277  0.00105 **
## Forgiveness.13           0.21876    0.06726   3.252  0.00114 **
## Gentleness.14           -0.51747    0.06897  -7.503 6.25e-14 ***
## Flexibility.15           0.23798    0.07871   3.024  0.00250 **
## Patience.16             -0.06097    0.06987  -0.873  0.38286
## Organization.17         -0.04182    0.08184  -0.511  0.60934
## Diligence.18             0.44723    0.08166   5.476 4.34e-08 ***
## Perfectionism.19        -0.47514    0.09474  -5.015 5.30e-07 ***
## Prudence.20             -0.45718    0.09683  -4.722 2.34e-06 ***
## AestheticApretiation.21 -0.41862    0.08344  -5.017 5.25e-07 ***
## Inquisitiveness22       -0.17628    0.08457  -2.084  0.03712 *
## Creativity.23            0.28737    0.06151   4.672 2.99e-06 ***
## Unconventionality.24    -0.55791    0.09212  -6.056 1.39e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4570.9  on 6219  degrees of freedom
## Residual deviance: 4194.3  on 6195  degrees of freedom
## AIC: 4244.3
##
## Number of Fisher Scoring iterations: 5
```

- 24 significant variables

```
####previsioni ####

resPP<-predict(modelPP, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano andati
in escursione
                                                    #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.5OptionPP, Predictedvalue= resPP>0.5)) #a 0,5 non predice valori T
RUE
```

	FALSE	TRUE
no	1363	4
si	181	7

```
(1372+7)/(1372+7+174+2) # 88,68%
```

```
## [1] 0.8868167
```

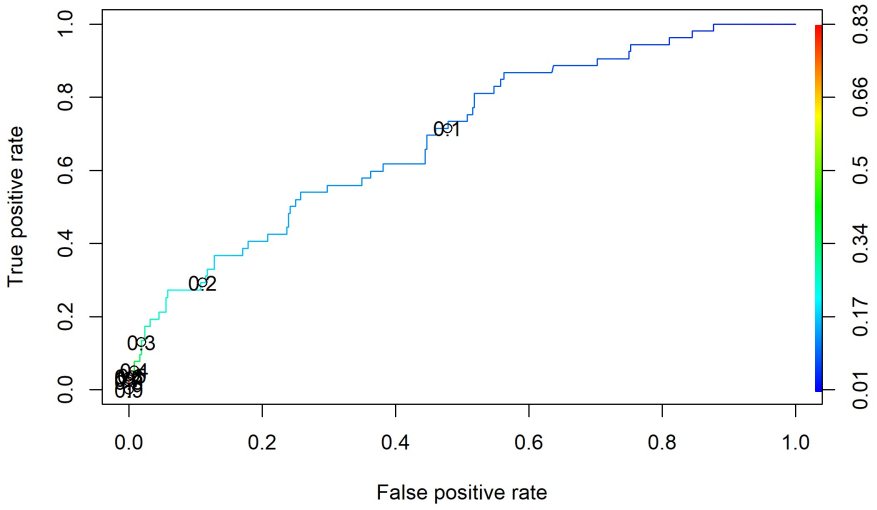
- The matrix show that model has an accuracy of 88.68%, more in details the matrix shows that the model perfectly predict when a person not choose PP as a political party that is fighting against climate change, but it fails completly in predicting when a person choose PP. Now i try to improve the prevision for the TRUE values

```
####migliorare la previsione####

res2PP<-predict(model2PP, training, type = "response") #predetto su dati training

ROCRPred.PP=prediction(res2PP, training$P43R43.5OptionPP)
ROCRPref.PP<-performance(ROCRPred.PP, "tpr", "fpr")

plot(ROCRPref.PP, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.2 come valore
```



- the graph suggest us to use the value of 0.2 instead of 0.5

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.5OptionPP, Predictedvalue= resPP>0.2))
```

	FALSE	TRUE
no	1217	150
si	133	55

```
(1227+54)/(1227+54+127+147) #accuratezza del 82,38%
```

```
## [1] 0.8237942
```

- Using the new value we can see that our model lose a little percentage of accuracy, however now the model can predict a little bit better when a person chose PP as a party that is fighting against climate change. The improvment it's very low. Now i want to graphically visualize my model.

```
####grafico delle probabilità predette ####

#create new table with data i need

dataPP<-data.frame( prob.of.vote= modelPP$fitted.values,
                    vote = training$P43R43.5OptionPP) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

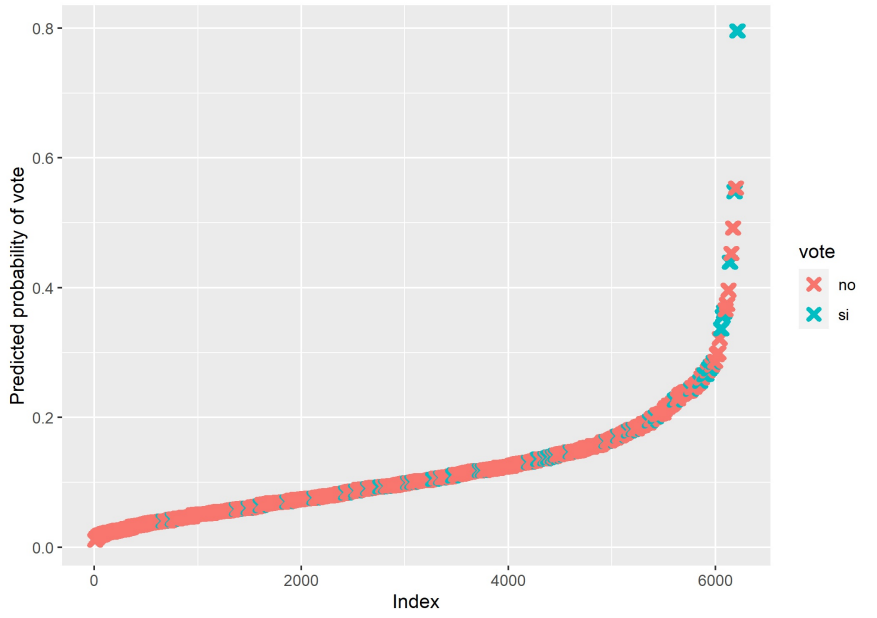
dataPP<-dataPP[order(dataPP$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataPP$rank<-1:nrow(dataPP)

#grafico

ggplot( data=dataPP ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.2, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability. These model can't be used to predict "green observation" but it's extremely precise in predicting when a person don't choose PP as a party that is fighting against climate change.

MAS PAIS

```
#summary backward
summary(modelMAS)
```

```
##
## Call:
## glm(formula = training$P43R43.6OptionMÃ.SPA ~ ., family = "binomial",
##     data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.0479  -0.7841  -0.5559   0.8351   2.5330
##
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## (Intercept)             -1.250020   0.035262 -35.450  < 2e-16 ***
## Sincerity.1              0.285198   0.066774   4.271 1.95e-05 ***
## Fairness.2              -0.157304   0.060269  -2.610  0.00905 **
## GreedAvoidance.3         0.255516   0.055589   4.597 4.30e-06 ***
## Modesty.4                0.296881   0.071238   4.167 3.08e-05 ***
## Fearfulness.5            0.255049   0.061408   4.153 3.28e-05 ***
## Anxiety.6               -0.139781   0.085668  -1.632  0.10275
## Dependence.7            -0.114423   0.064295  -1.780  0.07513 .
## Sentimentality.8        -0.159868   0.062831  -2.544  0.01095 *
## SocialSelf.Esteem.9      0.484004   0.066858   7.239 4.51e-13 ***
## SocialBoldness.10        0.329899   0.053201   6.201 5.61e-10 ***
## Sociability.11           0.336024   0.063373   5.302 1.14e-07 ***
## Liveliness.12           -0.499464   0.055896  -8.936  < 2e-16 ***
## Forgiveness.13          -0.148021   0.052531  -2.818  0.00484 **
## Gentleness.14            0.086094   0.052351   1.645  0.10006
## Flexibility.15          -0.870107   0.061031 -14.257  < 2e-16 ***
## Patience.16              0.669099   0.058341  11.469  < 2e-16 ***
## Organization.17         -0.491451   0.064322  -7.640 2.16e-14 ***
## Diligence.18            -0.343448   0.061986  -5.541 3.01e-08 ***
## Perfectionism.19        -0.132671   0.070594  -1.879  0.06019 .
## Prudence.20              0.845431   0.073425  11.514  < 2e-16 ***
## AestheticApretiation.21 -0.006448   0.067778  -0.095  0.92420
## Inquisitiveness22        0.347674   0.067032   5.187 2.14e-07 ***
## Creativity.23           -0.379145   0.047978  -7.902 2.74e-15 ***
## Unconventionality.24     0.363471   0.066145   5.495 3.91e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 7282.6  on 6219  degrees of freedom
## Residual deviance: 6288.4  on 6195  degrees of freedom
## AIC: 6338.4
##
## Number of Fisher Scoring iterations: 5
```

- 24 significant variables

```
####previsioni ####

resMAS<-predict(model2MAS, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano and
ati in escursione
                                                       #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.6OptionMÃ.SPA, Predictedvalue= resMAS>0.5)) #a 0,5 non predice val
ori TRUE
```

	FALSE	TRUE
no	1066	59
si	321	109

```
(1063+106)/(1063+106+324+62) # 75,18%
```

```
## [1] 0.7517685
```

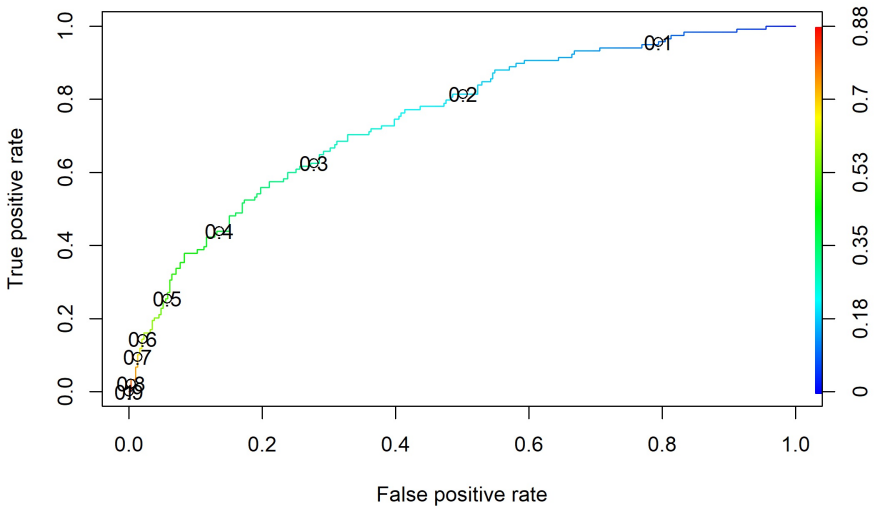
- The matrix show that model has an accuracy of 75.17%, more in details the matrix shows that the model can predict when a person not choose MAS PAIS as a political party that is fighting against climate change, but it fails 1/3 in predicting when a person choose MAS PAIS. Now i try to improve the prevision for the TRUE values

```
####migliorare la previsione####

res2MAS<-predict(model2MAS, training, type = "response") #predetto su dati training

ROCRPred.MAS=prediction(res2MAS, training$P43R43.6OptionMÃ.SPA)
ROCRPref.MAS<-performance(ROCRPred.MAS, "tpr", "fpr")

plot(ROCRPref.MAS, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #va bene 0,5
```



- the graph confirms the value of 0.5

```
####grafico delle probabilità predette ####

#create new table with data i need

dataMAS<-data.frame( prob.of.vote= model2MAS$fitted.values,
                     vote = training$P43R43.6OptionMÄ.SPA) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

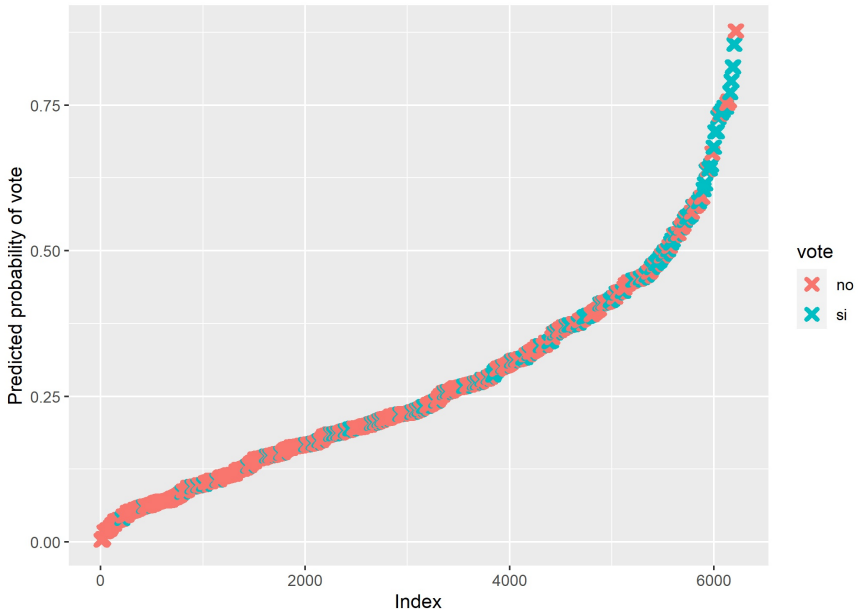
dataMAS<-dataMAS[order(dataMAS$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataMAS$rank<-1:nrow(dataMAS)

#grafico

ggplot( data=dataMAS ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.5, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability. These model can't be used to predict "green observation" but it's more precise in predicting when a person don't choose MAS PAIS as a party that is fighting against climate change.

CIUDADANOS

```
#summary backward
summary(model2CIU)
```

```
##
## Call:
## glm(formula = training$P43R43.8OptionCIUDA ~ GreedAvoidance.3 +
##     Fearfulness.5 + Anxiety.6 + Sentimentality.8 + SocialSelf.Esteem.9 +
##     SocialBoldness.10 + Sociability.11 + Forgiveness.13 + Flexibility.15 +
##     Organization.17 + Diligence.18 + Perfectionism.19 + Creativity.23 +
##     Unconventionality.24, family = "binomial", data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.0003  -0.4228  -0.3137  -0.2190   2.9823
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -2.84109    0.06235 -45.569  < 2e-16 ***
## GreedAvoidance.3    -0.66307    0.07631  -8.689  < 2e-16 ***
## Fearfulness.5       -0.52845    0.07708  -6.856 7.11e-12 ***
## Anxiety.6           -0.74662    0.10576  -7.060 1.67e-12 ***
## Sentimentality.8     0.57839    0.08073   7.165 7.80e-13 ***
## SocialSelf.Esteem.9 -0.43774    0.08741  -5.008 5.51e-07 ***
## SocialBoldness.10   -0.17439    0.06996  -2.493  0.01267 *
## Sociability.11       0.64618    0.07910   8.169 3.10e-16 ***
## Forgiveness.13      -0.21182    0.07346  -2.883  0.00394 **
## Flexibility.15       0.18340    0.08809   2.082  0.03734 *
## Organization.17     -0.56514    0.07836  -7.212 5.52e-13 ***
## Diligence.18         0.48874    0.09494   5.148 2.63e-07 ***
## Perfectionism.19    -0.16097    0.09837  -1.636  0.10176
## Creativity.23        0.45827    0.07138   6.420 1.36e-10 ***
## Unconventionality.24 -0.97264    0.09225 -10.543  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3341.4  on 6219  degrees of freedom
## Residual deviance: 2979.7  on 6205  degrees of freedom
## AIC: 3009.7
##
## Number of Fisher Scoring iterations: 6
```

- we have 14 significant variables

```
####previsioni ####

resCIU<-predict(model2CIU, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano and
ati in escursione
                                                       #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.8OptionCIUDA, Predictedvalue= resCIU>0.5)) #a 0,5
```

	FALSE	TRUE
no	1433	0
si	118	4

```
(1432+4)/(1432+4+119) # 92,34%
```

```
## [1] 0.9234727
```

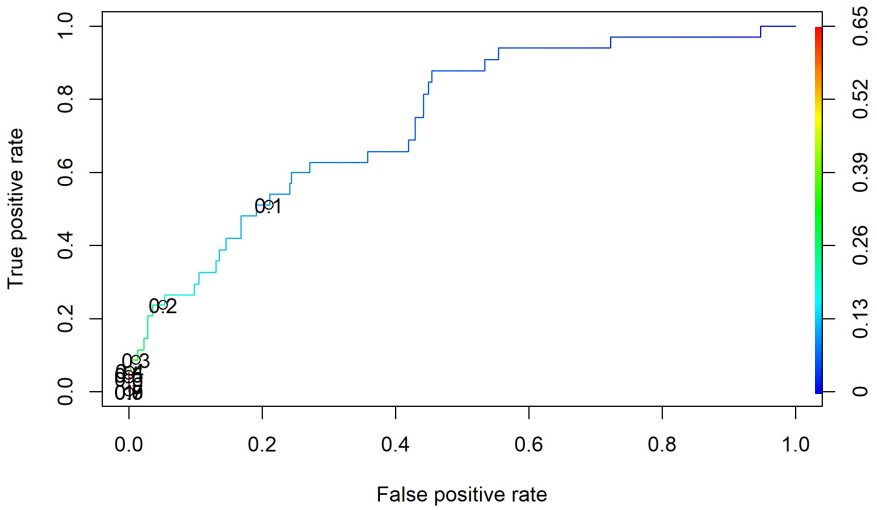
- The matrix show that model has an accuracy of 92.34%, more in details the matrix shows that the model can predict when a person not choose CIUDADANOS as a political party that is fighting against climate change, but it fails all the time in predicting when a person choose CIUDADANOS. Now i try to improve the prevision for the TRUE values

```
####migliorare la previsione####

res2CIU<-predict(model2CIU, training, type = "response") #predetto su dati training

ROCRPred.CIU=prediction(res2CIU, training$P43R43.8OptionCIUDA)
ROCRPref.CIU<-performance(ROCRPred.CIU, "tpr", "fpr")

plot(ROCRPref.CIU, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.2 come valore
```



- the graph suggest us we should use 0.2 iinstead of 0.5

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.8OptionCIUDA, Predictedvalue= resCIU>0.2))
```

	FALSE	TRUE
no	1364	69
si	90	32

```
(1362+33)/(1362+33+90+70) #accuratezza del 89,70%
```

```
## [1] 0.8971061
```

- Using the new value we can see that our model lose a little percentage of accuracy, however now the model can predict a little bit better when a person chose CIUDADANOS as a party that is fighting against climate change. The improvment it's very low. Now i want to graphically visualize my model.

```
####grafico delle probabilità predette ####

#create new table with data i need

dataCIU<-data.frame( prob.of.vote= modelCIU$fitted.values,
                     vote = training$P43R43.8OptionCIUDA) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

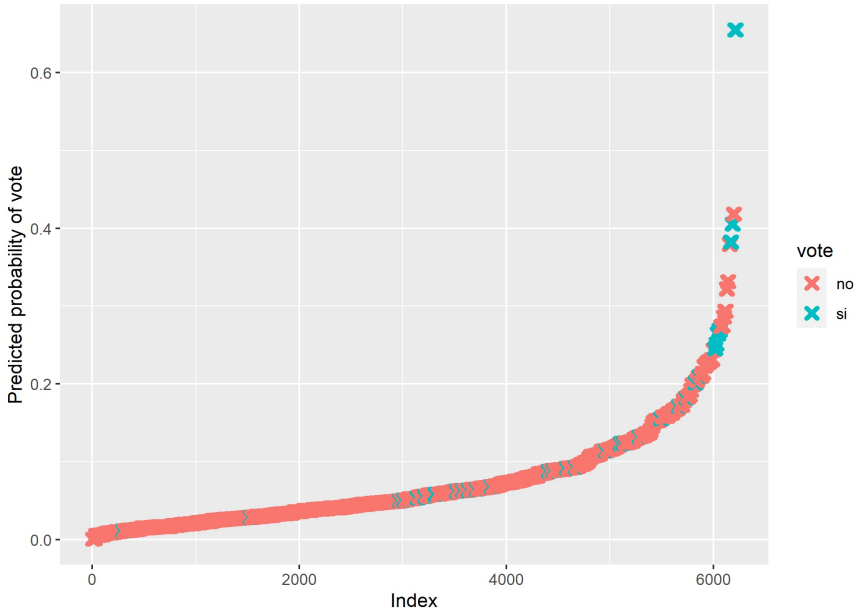
dataCIU<-dataCIU[order(dataCIU$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataCIU$rank<-1:nrow(dataCIU)

#grafico

ggplot( data=dataCIU ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.2, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability. These model can't be used to predict "green observation" but it's more precise in predicting when a person don't choose CIUDADANOS as a party that is fighting against climate change.

UPN

```
#summary backward
summary(model2UPN)
```

```
##
## Call:
## glm(formula = training$P43R43.12OptionUPN ~ Sincerity.1 + GreedAvoidance.3 +
##     Modesty.4 + Anxiety.6 + Dependence.7 + Sentimentality.8 +
##     SocialSelf.Esteem.9 + Liveliness.12 + Gentleness.14 + Flexibility.15 +
##     Patience.16 + Organization.17 + Diligence.18 + Perfectionism.19 +
##     Prudence.20 + AestheticApretiation.21 + Creativity.23 + Unconventionality.24,
##     family = "binomial", data = training)
##
## Deviance Residuals:
##      Min        1Q    Median        3Q       Max
## -0.91658  -0.10459  -0.03946  -0.01606   2.99689
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -6.8825     0.3492 -19.710  < 2e-16 ***
## Sincerity.1               1.5761     0.3128   5.038 4.70e-07 ***
## GreedAvoidance.3         -1.2881     0.2057  -6.263 3.79e-10 ***
## Modesty.4                -0.7955     0.2547  -3.123 0.001791 **
## Anxiety.6                -1.0469     0.2667  -3.926 8.65e-05 ***
## Dependence.7              1.3139     0.1899   6.920 4.51e-12 ***
## Sentimentality.8         -1.4296     0.2180  -6.557 5.51e-11 ***
## SocialSelf.Esteem.9       2.6601     0.2897   9.181  < 2e-16 ***
## Liveliness.12            -0.3126     0.1760  -1.776 0.075757 .
## Gentleness.14            -1.7453     0.2357  -7.405 1.31e-13 ***
## Flexibility.15            0.7148     0.1919   3.725 0.000195 ***
## Patience.16               1.9349     0.2536   7.631 2.33e-14 ***
## Organization.17          -1.1029     0.2128  -5.184 2.18e-07 ***
## Diligence.18              0.7041     0.2091   3.367 0.000760 ***
## Perfectionism.19          3.2175     0.3667   8.773  < 2e-16 ***
## Prudence.20              -2.7814     0.3599  -7.729 1.09e-14 ***
## AestheticApretiation.21  -0.4887     0.2287  -2.137 0.032602 *
## Creativity.23            -0.8095     0.1913  -4.232 2.31e-05 ***
## Unconventionality.24      1.6994     0.2643   6.430 1.27e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1040.87  on 6219  degrees of freedom
## Residual deviance:  622.18  on 6201  degrees of freedom
## AIC: 660.18
##
## Number of Fisher Scoring iterations: 9
```

- 19 significant variables

```
####previsioni ####

resUPN<-predict(model2UPN, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano and
ati in escursione
                                                #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.12OptionUPN, Predictedvalue= resUPN>0.5)) #a 0,5
```

	FALSE	TRUE
no	1531	0
si	20	4

```
(1527+3)/(1527+3+25) # 98,39%
```

```
## [1] 0.9839228
```

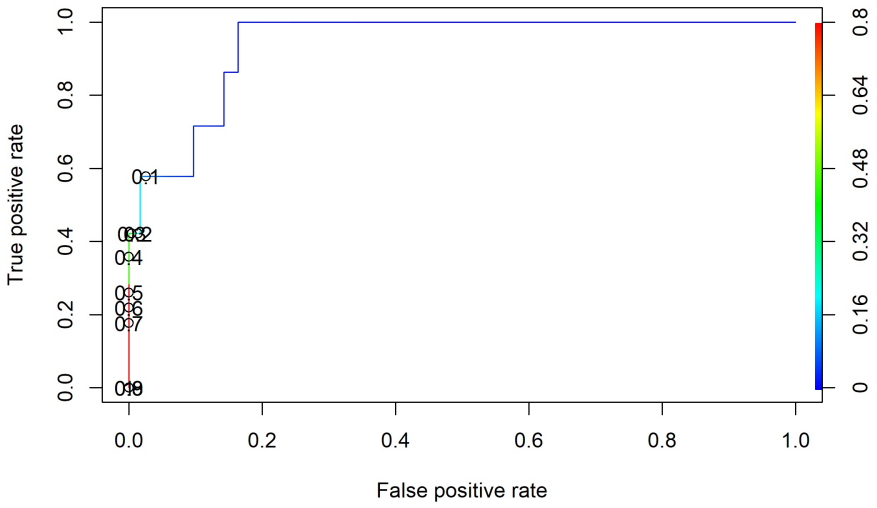
- The matrix show that model has an accuracy of 98.39%, more in details the matrix shows that the model can predict when a person not choose UPN as a political party that is fighting against climate change, but it fails all the time in predicting when a person choose UPN. Now i try to improve the prevision for the TRUE values

```
####migliorare la previsione####

res2UPN<-predict(model2UPN, training, type = "response") #predetto su dati training

ROCRPred.UPN=prediction(res2UPN, training$P43R43.12OptionUPN)
ROCRPref.UPN<-performance(ROCRPred.UPN, "tpr", "fpr")

plot(ROCRPref.UPN, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.1 come valore
```



- graph suggests us to use 0.1 instead of 0.5

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.12OptionUPN, Predictedvalue= resUPN>0.1))
```

	FALSE	TRUE
no	1494	37
si	11	13

```
(1508+11)/(1508+11+17+19) #accuratezza del 97,68%
```

```
## [1] 0.9768489
```

- Using the new value we can see that our model lose a little percentage of accuracy, however now the model can predict a little bit better when a person chose UPN as a party that is fighting against climate change. The improvment it's very low. Now i want to graphically visualize my model.

```
####grafico delle probabilità predette ####

#create new table with data i need

dataUPN<-data.frame( prob.of.vote= modelUPN$fitted.values,
                vote = training$P43R43.12OptionUPN) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

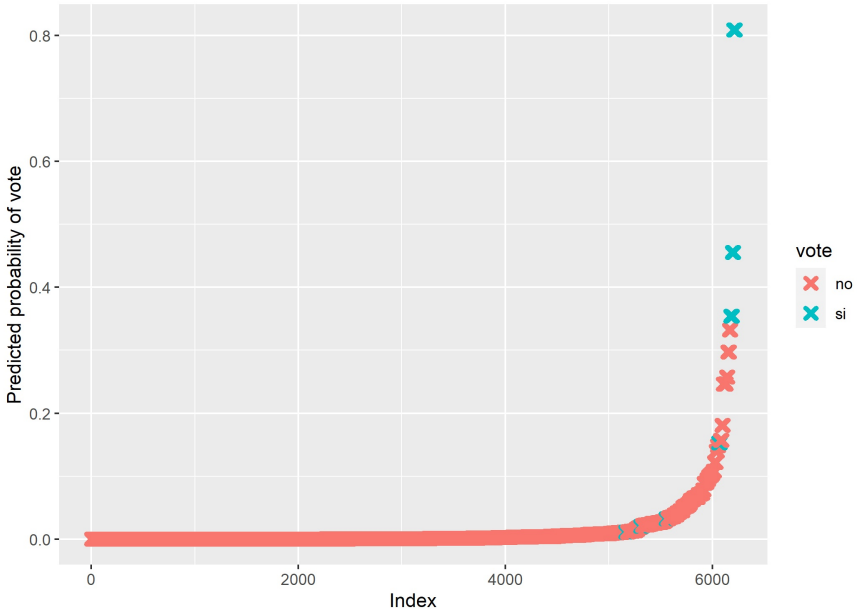
dataUPN<-dataUPN[order(dataUPN$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataUPN$rank<-1:nrow(dataUPN)

#grafico

ggplot( data=dataUPN ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.1, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability. These model can't be used to predict "green observation" but it's more precise in predicting when a person don't choose UPN as a party that is fighting against climate change.

COMPROMIS

```
#summary backward
summary(model2COM)
```

```
##
## Call:
## glm(formula = training$P43R43.13OptionCOMP ~ Sincerity.1 + Fairness.2 +
##     GreedAvoidance.3 + Modesty.4 + Anxiety.6 + SocialSelf.Esteem.9 +
##     SocialBoldness.10 + Sociability.11 + Liveliness.12 + Forgiveness.13 +
##     Gentleness.14 + Flexibility.15 + Patience.16 + Diligence.18 +
##     Perfectionism.19 + Prudence.20 + Inquisitiveness22 + Creativity.23 +
##     Unconventionality.24, family = "binomial", data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.6522  -0.5870  -0.4466  -0.2816   3.0740
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)          -2.00189    0.04477 -44.711  < 2e-16 ***
## Sincerity.1           0.25106    0.07454   3.368 0.000757 ***
## Fairness.2           -0.11250    0.06882  -1.635 0.102109
## GreedAvoidance.3     -0.18383    0.06405  -2.870 0.004106 **
## Modesty.4            -0.55476    0.07941  -6.986 2.83e-12 ***
## Anxiety.6             0.14533    0.07812   1.860 0.062847 .
## SocialSelf.Esteem.9   0.30200    0.07259   4.160 3.18e-05 ***
## SocialBoldness.10    -0.21391    0.05850  -3.657 0.000256 ***
## Sociability.11        0.12715    0.07344   1.731 0.083394 .
## Liveliness.12        -0.23593    0.06133  -3.847 0.000120 ***
## Forgiveness.13        0.16043    0.06307   2.543 0.010977 *
## Gentleness.14         0.17264    0.05865   2.943 0.003245 **
## Flexibility.15       -1.26182    0.07177 -17.581  < 2e-16 ***
## Patience.16           0.30110    0.06077   4.954 7.26e-07 ***
## Diligence.18          0.14665    0.07312   2.006 0.044886 *
## Perfectionism.19     -0.38779    0.08104  -4.785 1.71e-06 ***
## Prudence.20           0.51912    0.06991   7.426 1.12e-13 ***
## Inquisitiveness22     0.39657    0.07368   5.382 7.36e-08 ***
## Creativity.23        -0.50832    0.05535  -9.184  < 2e-16 ***
## Unconventionality.24  0.35543    0.07319   4.856 1.20e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 5538.0  on 6219  degrees of freedom
## Residual deviance: 4825.1  on 6200  degrees of freedom
## AIC: 4865.1
##
## Number of Fisher Scoring iterations: 5
```

- there are 19 significant variables

```
####previsioni ####

resCOM<-predict(model2COM, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano and
ati in escursione
                                                       #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.13OptionCOMP, Predictedvalue= resCOM>0.5)) #a 0,5
```

	FALSE	TRUE
no	1296	15
si	218	26

```
(1285+25)/(1285+25+228+17) # 84,24%
```

```
## [1] 0.8424437
```

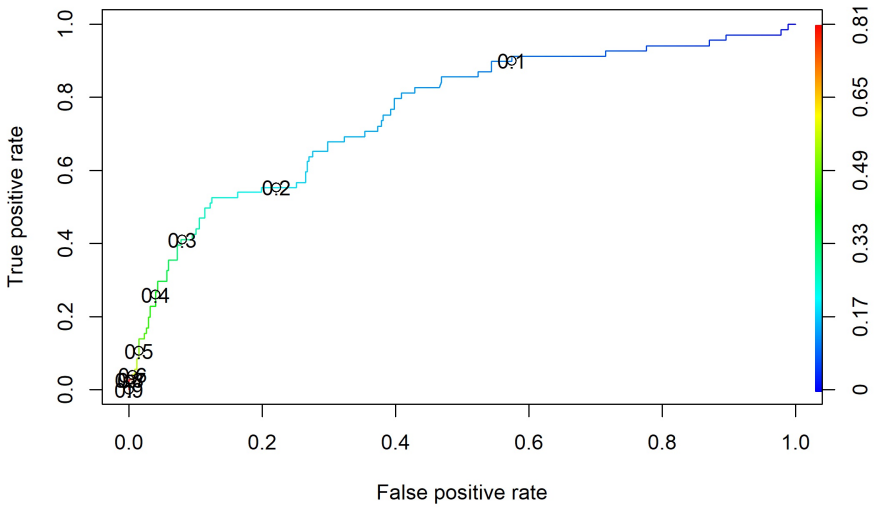
- The matrix show that model has an accuracy of 84.24%, more in details the matrix shows that the model can predict when a person not choose UPN as a political party that is fighting against climate change, but it fails all the time in predicting when a person choose COMPROMIS. Now i try to improve the prevision for the TRUE values

```
####migliorare la previsione####

res2COM<-predict(model2COM, training, type = "response") #predetto su dati training

ROCRPred.COM=prediction(res2COM, training$P43R43.13OptionCOMP)
ROCRPref.COM<-performance(ROCRPred.COM, "tpr", "fpr")

plot(ROCRPref.COM, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.3 come valore
```



- the graph suggests us to use 0.3 as value

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.13OptionCOMP, Predictedvalue= resCOM>0.3))
```

	FALSE	TRUE
no	1212	99
si	139	105

```
(1192+102)/(1192+102+152+109) #accuratezza del 83,21%
```

```
## [1] 0.8321543
```

- Using the new value we can see that our model lose a little percentage of accuracy, however now the model can predict a little bit better when a person chose COMPROMIS as a party that is fighting against climate change. The improvment it's very low. Now i want to graphically visualize my model.

```
####grafico delle probabilità predette ####

#create new table with data i need

dataCOM<-data.frame( prob.of.vote= modelCOM$fitted.values,
                vote = training$P43R43.13OptionCOMP) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

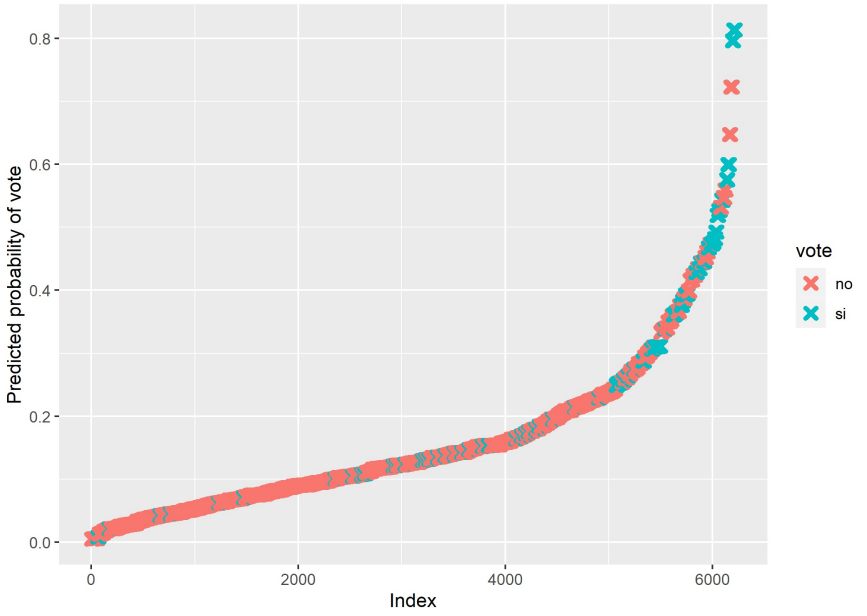
dataCOM<-dataCOM[order(dataCOM$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataCOM$rank<-1:nrow(dataCOM)

#grafico

ggplot( data=dataCOM ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.3, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability. These model can't be used to predict "green observation" but it's more precise in predicting when a person don't choose COMPROMIS as a party that is fighting against climate change.

PRC

```
#summary backward
summary(model2PRC)
```

```
##
## Call:
## glm(formula = training$P43R43.17OptionPRC ~ Sincerity.1 + GreedAvoidance.3 +
##     Modesty.4 + Fearfulness.5 + Anxiety.6 + Dependence.7 + SocialSelf.Esteem.9 +
##     SocialBoldness.10 + Sociability.11 + Liveliness.12 + Forgiveness.13 +
##     Flexibility.15 + Patience.16 + Organization.17 + Diligence.18 +
##     Prudence.20 + AestheticApretiation.21 + Inquisitiveness22 +
##     Creativity.23 + Unconventionality.24, family = "binomial",
##     data = training)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -0.8806  -0.2366  -0.1379  -0.0784   3.2589
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -4.6044     0.1498 -30.744  < 2e-16 ***
## Sincerity.1               0.5792     0.1678   3.451 0.000558 ***
## GreedAvoidance.3         -0.8165     0.1490  -5.479 4.28e-08 ***
## Modesty.4                -0.6593     0.1844  -3.574 0.000351 ***
## Fearfulness.5            -0.3279     0.1555  -2.109 0.034981 *
## Anxiety.6                 0.8028     0.2122   3.783 0.000155 ***
## Dependence.7              0.7235     0.1493   4.846 1.26e-06 ***
## SocialSelf.Esteem.9       1.4714     0.1747   8.422  < 2e-16 ***
## SocialBoldness.10         0.3303     0.1289   2.562 0.010394 *
## Sociability.11           -0.2579     0.1502  -1.717 0.085942 .
## Liveliness.12            -0.5652     0.1378  -4.101 4.12e-05 ***
## Forgiveness.13            0.7270     0.1561   4.656 3.22e-06 ***
## Flexibility.15           -1.3960     0.1775  -7.866 3.66e-15 ***
## Patience.16               0.7470     0.1422   5.254 1.49e-07 ***
## Organization.17          -0.9326     0.1611  -5.789 7.06e-09 ***
## Diligence.18              0.3278     0.1640   1.998 0.045715 *
## Prudence.20               0.4348     0.1662   2.617 0.008874 **
## AestheticApretiation.21   0.4436     0.1625   2.729 0.006348 **
## Inquisitiveness22        -0.9372     0.1757  -5.333 9.65e-08 ***
## Creativity.23             0.3812     0.1141   3.342 0.000833 ***
## Unconventionality.24     -0.2267     0.1477  -1.534 0.124952
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1573.5  on 6219  degrees of freedom
## Residual deviance: 1256.1  on 6199  degrees of freedom
## AIC: 1298.1
##
## Number of Fisher Scoring iterations: 8
```

- the model has 20 significant variables

```
####previsioni ####

resPRC<-predict(model2PRC, testing, type = "response") #abbiamo le prob per le osservazioni testing che siano and
ati in escursione
                                                       #modello predetto su dati testing
####matrice accuratezza del modello####

knitr::kable(table(Actualvalue= testing$P43R43.17OptionPRC, Predictedvalue= resPRC>0.5)) #a 0,5
```

	FALSE	TRUE
no	1511	0
si	40	4

```
(1511+3)/(1511+3+41) # 97,36%
```

```
## [1] 0.9736334
```

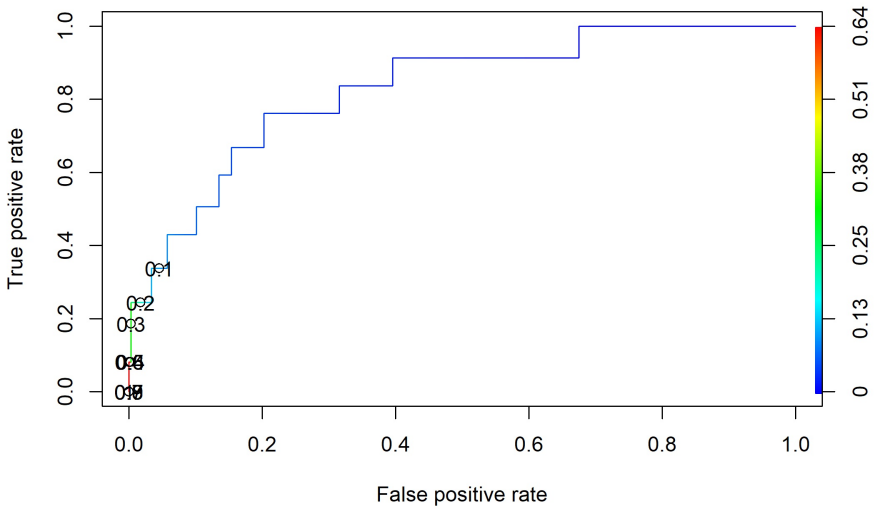
- The matrix show that model has an accuracy of 97.36%, more in details the matrix shows that the model can predict when a person not choose UPN as a political party that is fighting against climate change, but it fails all the time in predicting when a person choose PRC. Now i try to improve the prevision for the TRUE values

```
####migliorare la previsione####

res2PRC<-predict(model2PRC, training, type = "response") #predetto su dati training

ROCRPred.PRC=prediction(res2PRC, training$P43R43.17OptionPRC)
ROCRPref.PRC<-performance(ROCRPred.PRC, "tpr", "fpr")

plot(ROCRPref.PRC, colorize=TRUE, print.cutoffs.at=seq(0.1, by=0.1)) #ci suggerisce di usare 0.1 come valore
```



- the grapgh suggests to us to use 0.1 as value

```
#ritestiamo il modello

knitr::kable(table(Actualvalue= testing$P43R43.17OptionPRC, Predictedvalue= resPRC>0.1))
```

	FALSE	TRUE
no	1442	69
si	30	14

```
(1445+10)/(1445+10+34+66) #accuratezza del 93,57%
```

```
## [1] 0.9356913
```

- Using the new value we can see that our model lose a little percentage of accuracy, however now the model can predict a little bit better when a person chose PRC as a party that is fighting against climate change. The improvment it's very low. Now i want to graphically visualize my model.

```
#grafico delle probabilità predette #

#create new table with data i need

dataPRC<-data.frame( prob.of.vote= modelPRC$fitted.values,
                vote = training$P43R43.17OptionPRC) #ho usato training pk così avevano stesso numero colonne

#ordino in prob crescente

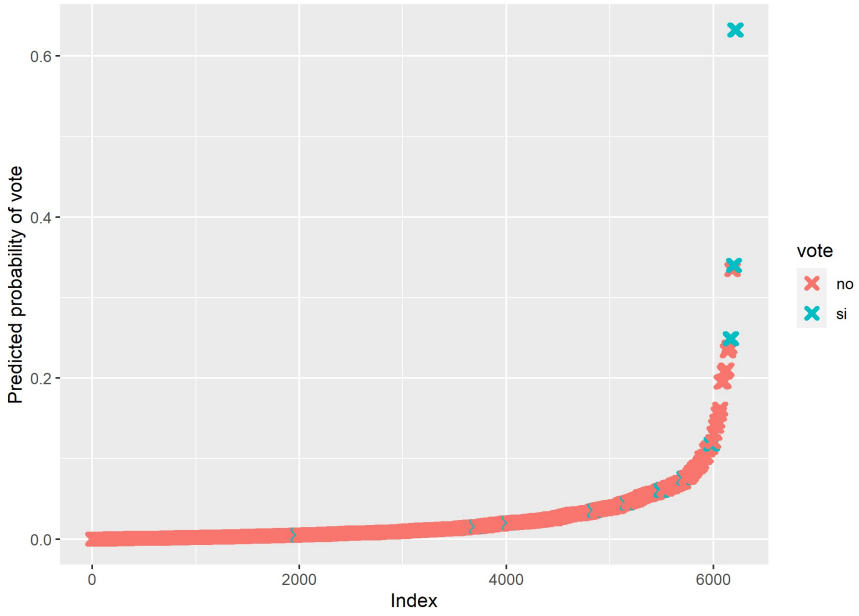
dataPRC<-dataPRC[order(dataPRC$prob.of.vote , decreasing = F),]

#creo nuova colonna con prob ordinate

dataPRC$rank<-1:nrow(dataPRC)

#grafico

ggplot( data=dataPRC ,
        aes(x=rank, y= prob.of.vote))+
  geom_point(aes(color=vote), alpha=1 , shape=4, stroke=2)+
  xlab("Index")+
  ylab("Predicted probability of vote")
```



- From the graph we can see how the observation are distributed second their probability. we can see that the most of the red observations are under the values of 0.1, instead we can see that our model fail to predict green observations, in fact we can see that lots of these observations have a low values of probability. These model can't be used to predict "green observation" but it's more precise in predicting when a person don't choose PRC as a party that is fighting against climate change.

Conclusion 2

All this model give us a probability to choose or not to choose a political party that people believe it could fight the climate change due to the personality traits, we can see how all these models are able to better predict when people don't choose the party than when they do it. Even if trying to improve the models this aspect doesn't change. This is also given because lots of the coefficient in the models are negatives so most of the time increasing the independents variables ends up in decreasing the probability of Y=1.

3) PRINCIPAL COMPONENTS ANALYSIS

The second part of this investigation will focus on the principal components analysis. I want to see if there are principal components able to explain the correlation between personality traits and the political parties chosen by people answering the question: "Cuál de los siguientes partidos políticos cree que apoya la lucha contra el cambio climático?

How i will proceed:

1. to run the PCA i will use the table of the correlation between personality traits and political parties, in this way i will have a table in which all the data are in the same dimension. i will manipulate the table in order to have on the Y the political parties and on the X the traits of personalities.
2. i run the PCA with the "prcomp" command, then i analyze the summary of the PCA model studying the scores of the principal components searching for patterns
3. i graphically visualize the results of the PCA with the biplot graph, i will create a new categorical variable called "ideology" to color differently the izquierda and derecha parties. in this way should be easier reading data and find correlation with the principal components
4. i create a correlation matrix between the variables and the scores of the principal components, this should be very helpful in interpreting data together with the prior graph.

```
mat2<-mat0[-c(12:35), -c(1:11)]

data<-as.data.frame(mat2) # crato dataframe da matrice

data$ideologia<-c( "derecha", "izquierda", "izquierda", "derecha",
                   "derecha", "izquierda", "izquierda", "derecha", "derecha", "izquierda", "izquierda")

personalities.pca<-prcomp(data[,-25], scale. = F)

summary(personalities.pca) # PC1 explain 42.75% of variance
```

```
## Importance of components:
##                           PC1    PC2     PC3     PC4     PC5     PC6     PC7
## Standard deviation     0.1747 0.1345 0.08372 0.07301 0.06429 0.04938 0.04256
## Proportion of Variance 0.4275 0.2534 0.09820 0.07468 0.05790 0.03417 0.02538
## Cumulative Proportion  0.4275 0.6809 0.77908 0.85375 0.91165 0.94582 0.97120
##                           PC8     PC9    PC10      PC11
## Standard deviation     0.02845 0.02571 0.02419 1.571e-17
## Proportion of Variance 0.01134 0.00926 0.00820 0.000e+00
## Cumulative Proportion  0.98254 0.99180 1.00000 1.000e+00
```

```
                          # PC2 explain 25.34% of Variance; together the first two explain 68.09% of variance
                          # we need until PC4 to explain at list 85% of variance
knitr::kable(personalities.pca$rotation) #we can see the scores of PC for each variables
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11
Sincerity.1	0.0913569	0.1643407	-0.2006724	-0.2843581	0.0353971	-0.2808008	-0.1028836	-0.1937189	-0.1828291	-0.0070117	-0.0616002
Fairness.2	0.1277526	0.1703988	-0.1156321	-0.0229376	0.2236371	-0.0367482	-0.1317561	-0.2702560	-0.1196304	-0.2488155	0.7205050
GreedAvoidance.3	0.1603397	-0.0912911	-0.0745197	-0.1693244	0.4599245	-0.2448760	0.1446753	0.2769129	-0.4715984	-0.0144200	-0.0545379
Modesty.4	0.2471086	0.1741014	0.0795363	-0.3481057	0.0461983	0.0269401	-0.3387503	-0.2739852	-0.1265763	0.0436321	-0.3601297
Fearfulness.5	-0.1470399	-0.0763149	0.3835153	-0.0065815	0.0864691	-0.1006127	-0.4136638	0.2582025	0.0660269	-0.0511907	0.1046083
Anxiety.6	-0.0249923	0.2087063	0.2530420	0.2838432	0.2365222	-0.1936003	-0.0168607	-0.1356902	-0.0588682	0.2158532	-0.1448588
Dependence.7	-0.0709736	0.2506999	0.4923018	0.1178811	-0.1034441	-0.0317671	0.0779231	0.0121241	-0.1317746	-0.0359955	-0.0611416
Sentimentality.8	-0.0395279	0.2242467	0.3749325	-0.1067143	0.1542737	0.2307140	-0.0120109	-0.0226090	0.1248856	-0.0459964	0.0303075
SocialSelf.Esteem.9	0.0003434	0.3603952	-0.1347859	-0.1256268	-0.0697493	0.1235491	0.1153509	-0.1400240	-0.1952573	0.1907019	-0.0928026
SocialBoldness.10	0.0410698	-0.0547371	-0.2119358	0.0265108	0.2844907	0.3609200	0.0981070	0.0052187	0.0810714	0.5575399	0.0452140
Sociability.11	0.0205835	-0.3177425	0.1802346	-0.2521143	0.0544697	-0.0099581	-0.0438018	-0.0546894	0.0821834	0.2869878	0.2616749
Liveliness.12	-0.0783095	0.1170858	-0.1442960	-0.2174348	0.0831032	-0.0006042	0.0876339	0.3921149	0.0912390	-0.0634311	0.0435464
Forgiveness.13	0.0477382	-0.0221170	-0.0408070	-0.1510651	0.2342479	-0.2099669	0.2236300	0.1190499	0.2858867	-0.4408583	-0.2122102
Gentleness.14	0.0479496	-0.0853498	0.1909802	-0.3673061	-0.2660199	0.1080990	0.4188095	-0.0522927	-0.0559545	0.0176975	0.1380949
Flexibility.15	-0.2732964	0.4218541	-0.0711000	-0.2364289	0.2648702	-0.1528278	-0.0358200	-0.0182431	0.5030004	0.1912858	0.0140655
Patience.16	0.2892451	0.2100312	0.1011485	-0.2792779	-0.3279020	-0.1348999	-0.0810095	0.3564116	0.0675753	0.1033836	0.2045946

Organization.17	0.0796633	0.1937652	-0.2193793	0.1715821	-0.1452337	0.0550238	-0.0718956	0.3977713	-0.0560155	0.0416673	-0.1776615
Diligence.18	-0.1156757	0.3594803	-0.1066698	0.3238876	-0.0141071	-0.0298213	0.2621236	-0.0292413	-0.0330660	-0.1440520	0.2029366
Perfectionism.19	-0.2183931	0.1362808	0.1039472	-0.1028244	-0.2053376	-0.2580782	0.3528238	0.0257597	-0.1912563	0.2263395	0.0646833
Prudence.20	0.4863425	0.1815065	0.1176128	0.1937785	-0.0060694	0.0202636	-0.0589699	-0.0007439	-0.0908617	0.0244005	0.0111355
AestheticApretiation.21	0.2971043	0.1197354	-0.0223676	-0.1027348	-0.0761410	0.4091550	0.1229177	-0.0966396	0.2321204	-0.2642273	-0.0862493
Inquisitiveness22	0.1872147	-0.1630870	0.1973687	0.0402279	0.1855743	-0.1899821	0.3854881	-0.2482544	0.1550859	-0.0120435	-0.1398413
Creativity.23	-0.1822127	0.0760248	0.1892587	-0.1238577	0.3276581	0.4845177	0.0894937	0.2077516	-0.3175168	-0.1468927	-0.0054783
Unconventionality.24	0.4775653	0.0197964	0.1058261	0.1842231	0.1645388	-0.0715898	0.1393161	0.2396407	0.2074142	0.2047038	0.1286869

```
knitr::kable(personalities.pca$x)        #scores for PC values for the single observation
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11
P43R43.1OptionVOX	-0.2272834	-0.0805322	-0.0813517	0.0739200	0.0573978	0.0193691	-0.0404807	0.0357833	-0.0100085	0.0294407	0
P43R43.2OptionPODEM	0.1848548	-0.0842574	0.0273274	-0.0573653	0.0066368	-0.0038282	-0.0500357	0.0336454	0.0495731	-0.0141254	0
P43R43.3OptionIZQUI	0.1721247	-0.0300291	0.1289552	0.0349550	0.1275040	-0.0128982	0.0271273	-0.0186705	-0.0063763	0.0182316	0
P43R43.4OptionPNV	-0.0329267	0.0485948	-0.0222803	-0.0664004	-0.0010206	-0.0346990	0.0871396	0.0490209	-0.0131867	-0.0080498	0
P43R43.5OptionPP	-0.1951534	0.0064665	-0.0433764	-0.0030271	0.0620214	-0.0507815	-0.0214051	-0.0291212	-0.0043333	-0.0498824	0
P43R43.6OptionMÁ.SPA	0.2217809	-0.0559634	-0.0431510	-0.0845803	-0.0347753	0.0032258	-0.0442957	-0.0087657	-0.0526288	0.0044579	0
P43R43.7OptionBILDU	0.1606328	0.0399454	-0.1750952	0.0286469	-0.0056707	0.0084742	0.0374020	-0.0325851	0.0274565	0.0146809	0
P43R43.8OptionCIUDA	-0.2281195	-0.1596254	0.0551555	-0.0854817	-0.0330556	0.0772595	0.0275910	-0.0260859	0.0099384	0.0054016	0
P43R43.12OptionUPN	-0.1397070	0.1758565	0.0622602	-0.0278801	-0.0659607	-0.0753987	-0.0260909	-0.0138209	0.0113365	0.0329224	0
P43R43.13OptionCOMP	0.0567153	-0.1460069	0.0526524	0.1452026	-0.1055518	-0.0197339	0.0153127	0.0022872	-0.0063960	-0.0153466	0
P43R43.17OptionPRC	0.0270814	0.2855512	0.0389039	0.0420104	-0.0075254	0.0890110	-0.0122644	0.0083124	-0.0053750	-0.0177309	0

- from the summary of the PCA we can see how PC1 explain 42.75% of the variance and PC2 explain 25.34%, together the first two components explain 68.09% of variance, we can consider this a good result to proceed in our analysis
- the first table show the PC scores for each variable, at a first look i can't say if there is a pattern or a correlation among the PC scores, we have to wait for further investigation
- the second table show the PC scores for each political party, in this case for PC1 we can notice that political parties that belongs to the derecha have a negative score, instead the ones that belongs to the izquierda have a positive score

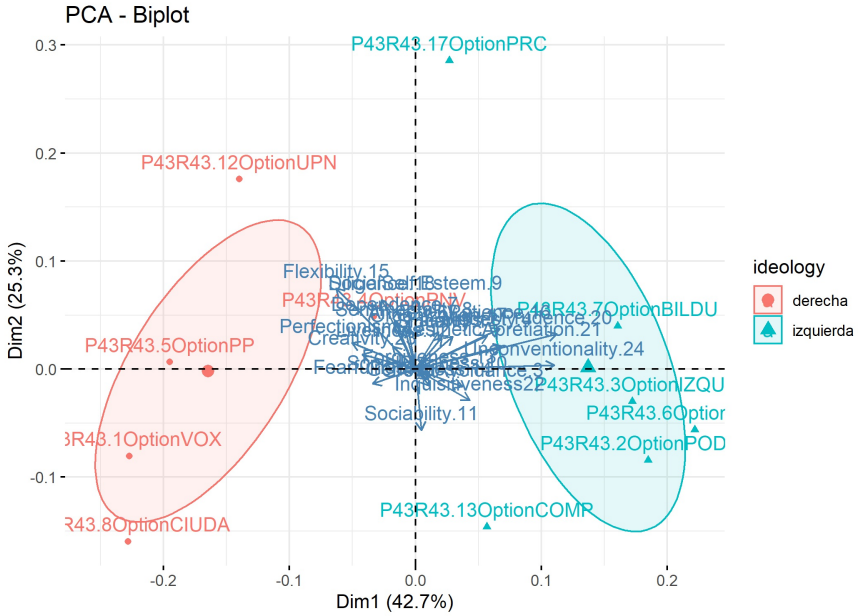
Now i want to visualize the results of this 2 tables on a graph. i will use the biplot graph, on the x i'll have PC1 and on the y PC2. Variables and political parties will be distributed as a combination of their scores in PC1 and PC2.

```
### grafico

#biplot

ideology<-as.factor(data$ideologia)

fviz_pca_biplot(personalities.pca,
                col.ind = ideology,
                addEllipses = TRUE,
                ellipse.type= "confidence",
                legend.title= "ideology",
                repel= F,
                )
```



- For what concern Political parties the graph show an evident split between derecha (red) and izquierda (blu) parties explained by the scores of PC1, the new categorical variable "ideology" together with the ellipses help us in visualizing better that pattern.
- For what concern the Personality Traits the graph shows us that some traits are more connected than other to the choice of a political party that people believe is fighting against climate change. In particular traits like prudence, unconventionality, Aesthetic Appreciation and Modesty are more connected to izquierda parties.

However i want to statistically prove that this correlation between some Personality Traits and Political Parties exists. To do this i will extract the PC1 & 2 scores and i'll run a correlation matrix with the initial data, what i expect is that traits of personality that on the graph are closer to izquierda parties have a higher positive correlation number in PC1, and the one closer to derecha parties have a higher negative correlation number in PC1.

```
###estrazione componenti principali e correlazione con le variabili

data2<-cbind(data , personalities.pca$x[,1:2]) #creo nuovo dataframe con i valori

knitr::kable(cor(data[,-25], data2[,26:27])) #vediamo le correlazioni tra le variabili e le componenti principali
```

	PC1	PC2
Sincerity.1	0.3841539	0.5320703
Fairness.2	0.5828343	0.5985519
GreedAvoidance.3	0.5723585	-0.2509086

Modesty.4	0.7410616	0.4020023
Fearfulness.5	-0.5450884	-0.2178215
Anxiety.6	-0.0963543	0.6195260
Dependence.7	-0.2214709	0.6023288
Sentimentality.8	-0.1457654	0.6367016
SocialSelf.Esteem.9	0.0011534	0.9320110
SocialBoldness.10	0.2005128	-0.2057599
Sociability.11	0.0722476	-0.8586937
Liveliness.12	-0.4308788	0.4960263
Forgiveness.13	0.2919518	-0.1041432
Gentleness.14	0.1963343	-0.2690751
Flexibility.15	-0.5981597	0.7108949
Patience.16	0.7573236	0.4234079
Organization.17	0.3476507	0.6510592
Diligence.18	-0.3402961	0.8142351
Perfectionism.19	-0.7568996	0.3636586
Prudence.20	0.9425679	0.2708459
AestheticApretiation.21	0.8702848	0.2700442
Inquisitiveness22	0.6678546	-0.4479411
Creativity.23	-0.6247627	0.2007020
Unconventionality.24	0.9642552	0.0307756

Our prevision are confirmed, in particular we have:

- PC1: positive correlation with Modesty, Patiente, Prudence, AestheticApretiation, Unconventionality and Inquisitiveness; negative correlation with Perfectionism and creativity.
- PC2: positive correlation with SocialSelf.Esteem, Flexibility, Diligence and Organization; negative correlation Sociability

Finally, I want to see if what we have found with these analysis can be proved by the table data. I take as a reference the variable Prudence which is the one most correlated to the left parties. I want to see if for high values of Prudence there are more people that believe these parties are fighting the climate change, to do so I will count the votes between the first and the last quantile of prudence. I expect more votes in the latest quantile to confirm my model.

```
### test

q<-quantile(y1$Prudence.20) #1 quantile -3.94 a -0.46 #4 quantile 0.57 a 4.93

# quantile 1

ytest<-y1[y1$Prudence.20< - 0.46,]        #restituisce tabella con quelle specifiche
                                         # e per quelle variabili
ytest<-ytest[,-c(12:30, 32:35)]

a<-table(ytest$P43R43.2OptionPODEM) #774
b<-table(ytest$P43R43.3OptionIZQUI) #594 #per valori bassi abbiamo meno voti nei partiti di sinistra
c<-table(ytest$P43R43.6OptionMÃ.SPA)#468
d<-table(ytest$P43R43.7OptionBILDU) #126

a<-774
b<-594
c<-468
d<-126

quantile1<-data.frame(parties = c("PODEMOS", "IZQUI", "MAS PAIS", "BILDU"), values=c(a,b,c,d))
knitr::kable(quantile1)
```

parties	values
PODEMOS	774
IZQUI	594
MAS PAIS	468
BILDU	126

```
#quantile 4

ytest1<-y1[y1$Prudence.20>0.57,]
ytest1<-ytest1[,-c(12:30, 32:35)]


e<-table(ytest1$P43R43.2OptionPODEM) #1079
f<-table(ytest1$P43R43.3OptionIZQUI) #972 #per valori alti abbiamo più voti nei partiti di sinistra
g<-table(ytest1$P43R43.6OptionMÃ.SPA)#683
h<-table(ytest1$P43R43.7OptionBILDU) #270

e<-1079
f<-972
g<-683
h<-270

quantile4<-data.frame(parties =c("PODEMOS", "IZQUI", "MAS PAIS", "BILDU"), values = c(e,f,g,h))
knitr::kable(quantile4)
```

parties	values
PODEMOS	1079
IZQUI	972
MAS PAIS	683
BILDU	270

As expected an higher score in Prudence increase the probability of choosing an izquierda parties as a party people believe is fighting against climate change.

Conclusion 3

The PCA shows great results, in particular it's very interesting the relations between the observations (x (political parties)) and PC1 in which we can recognize a clear pattern where derecha parties have negative scores and izquierda parties have positive scores. Furthermore, it's remarkable the correlations between the variables (personality traits) and the scores of PC1 & PC2 that actually proved that some traits of personality are more connected to other in the choice of a political party as a party that people believe is fighting against climate change.